The empirical recurrence for OEIS A231909, proved by transfer matrix

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Abstract

OEIS A231909 counts arrays of width 2 over $\{0, 1, 2\}$ in which no cell is beaten by a strict majority of its neighbours. The entry carries an empirical recurrence of order 6 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical: what a cell asks reaches one row above it and one row below, so the condition on a whole row is settled by 3 consecutive rows and the count is a walk count on $S = 85$ states.

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1 The conjecture

OEIS A231909 is "Number of 2 X n 0..2 arrays with no element having a strict majority of its horizontal, diagonal and antidiagonal neighbors equal to itself plus one mod 3, with upper left element zero (rock paper and scissors drawn positions).". It has offset 1 and begins

$$3, 17, 137, 948, 6975, 50323, 366170, 2657785, \ \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 9*a(n-1) - 5*a(n-2) - 66*a(n-3) + 78*a(n-4) + 4*a(n-5) - 16*a(n-6).

As of the "Last modified" line on the live entry (Oct 01 2018 09:14:13, revision 8) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The arrays have 2 columns and a growing number of rows, with entries in $\{0, 1, 2\}$. Fix the list of offsets the entry names,

$$\Delta = \{\,(-1,-1),(-1,0),(-1,1),(1,-1),(1,0),(1,1)\,\},$$

and for a cell (i, j) write

$$S(i,j) = \{\,(i+\delta_1,\ j+\delta_2) : (\delta_1,\delta_2) \in \Delta\,\} \cap (\text{the cells of the array})$$

for the neighbours it actually has. Only offsets landing inside the array count, so a cell on the border simply has fewer neighbours. The entry writes the array with n as the number of COLUMNS; transposing it exchanges the two coordinates of every offset, and the offsets above are already written in the transposed frame.

The condition is that no cell has a strict majority of its neighbours one greater than itself modulo 3:

$$2 \cdot \#\{(a,b) \in S(i,j) : x(a,b) \equiv x(i,j) + 1 \bmod 3\} \;\le\; |S(i,j)|.$$

With three values this is the drawn position of rock, paper and scissors: each value beats the one below it modulo 3, and no cell is to be beaten by a strict majority of its neighbours. The entry adds that the top-left cell is 0.

3 A window of 3 rows

The offsets reach one row up and one row down, so everything the condition asks of one row is decided by that row together with one row above it and one row below — a window of 3 consecutive rows, and nothing wider. Take the array one column at a time and let the state be the last 2 of them, with a distinguished start standing for the array before its first row, so that a walk of n steps is an array of n rows.

Appending a row completes exactly one window, and the row it settles is the one just above the bottom of that window; a step is allowed when that row passes. Each row of the finished array is therefore tested exactly once.

When the array stops, the last row of it has never been tested, because the conditions reach one row below. So the end vector is NOT all ones: a state is accepted exactly when those rows pass with nothing below them, which is decided by running the same test on the window with absent rows appended — the same rule that governs a cell at the bottom edge in Section 2. Thus

$$a(n) \;=\; \iota^{\top} M^{\,j} \tau, \qquad S = 85,$$

for the transfer matrix M on those S states. The walk index j and the entry's own n differ by a constant, read off by matching the published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^6 - \sum_i c_i t^{6-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M)\tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M)\tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+6} - \sum_i c_i M^{j+6-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M)\tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 9\,a(n-1) - 5\,a(n-2) - 66\,a(n-3) + 78\,a(n-4) + 4\,a(n-5) - 16\,a(n-6)$$

for every $n > 6$.

Proof. The vector $w = q(M)\tau$ was formed by 6 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 6 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Four checks, each independent of the algebra above.

First, the model reproduces all 20 terms the entry publishes, exactly, in integer arithmetic.

Second, the count was recomputed for the smallest arrays straight from the definition: every array of width 2 over $\{0, 1, 2\}$ with a few rows was written out cell by cell and each cell tested against the neighbours it has. No window, no state and no end vector are involved in that computation, and the published terms came back.

Third, the conjectured recurrence was evaluated directly on the published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Fourth, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A231909.

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[3] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.